

Simulation-Based Inference

Frequentist Simulation Studies

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Simulation Studies: Overview

- Choose a model $M^* = M^*(\theta, \psi)$ that represents the ground truth.
- Choose true values for θ and ψ , or a grid of such values.
- Choose the number of simulation trials S .
- For each choice of true parameters, simulate S datasets $y^{(s)} \sim M^*(\theta, \psi)$.
- Compute the estimates $\hat{\theta}^{(s)} = \hat{\theta}(y^{(s)})$, or more generally, the value of the test statistic of interest $T^{(s)} = T(y^{(s)})$.
- Compute further relevant statistics such as p-values, confidence intervals etc.
- Summarize the metrics across the the S trials appropriately.

Estimation Accuracy

Suppose we want to evaluate the accuracy of estimator $\hat{\theta}$ for parameter θ under model M^* .

Based on S simulated estimates $\hat{\theta}^{(s)}$, we can compute accuracy metrics such as the bias

$$\text{bias}(\hat{\theta}) = \frac{1}{S} \sum_{s=1}^S \hat{\theta}^{(s)} - \theta,$$

mean absolute error (MAE):

$$\text{MAE}(\hat{\theta}) = \frac{1}{S} \sum_{s=1}^S |\hat{\theta}^{(s)} - \theta|,$$

or the root mean squared error (RMSE):

$$\text{RMSE}(\hat{\theta}) = \sqrt{\frac{1}{S} \sum_{s=1}^S (\hat{\theta}^{(s)} - \theta)^2}$$

Example: Difference of two lognormal means

Setting: Our true model M^* consists of two submodels M_1 and M_2 .

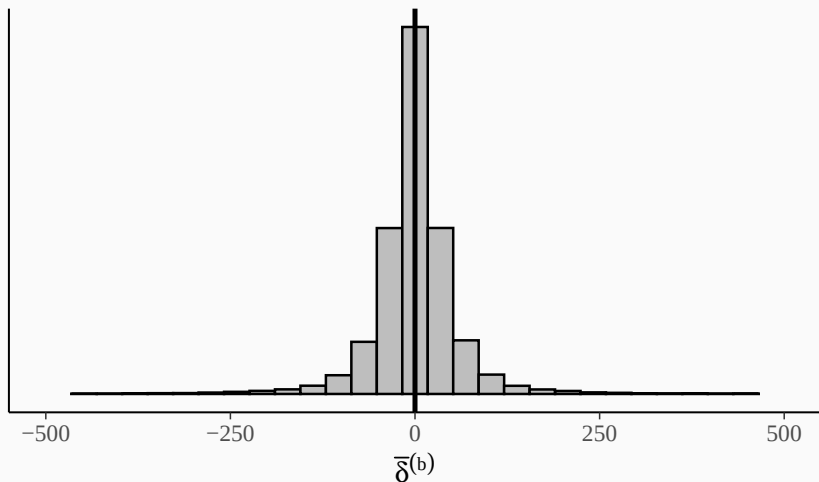
For $j = 1, 2$, $N = 40$, and, $i = 1, \dots, N$, we have

$$y_{ji} \sim M_j = \text{lognormal}(\mu = 2, \sigma = 2)$$

Goal: Estimate the true mean difference $\delta = 0$ via the estimator

$$\hat{\delta} = \bar{y}_1 - \bar{y}_2$$

Difference of lognormal means: Sampling distribution



Bias = 0.33, MAE = 41.75, RMSE = 146.77.

The sampling distribution of a p-value is also called *p-curve*.

For each simulation trial s , compute the p-value $p^{(s)}$ corresponding to the test-statistic $T^{(s)}$.

Plot the histogram of the set of S p-values $\{p^{(s)}\}_{s=1}^S$

Let M_0 be the assumed model of the statistical test.

Question: If $M_0 = M^*$, what is the sampling distribution of $p^{(s)}$?

Uniformity of p-curves

If $M_0 = M^*$, the sampling distribution of $p^{(s)}$ is (discretely) uniform.

Proof Sketch: If $M_0 = M^*$, then the sampling distribution $p_0(T)$ of T assumed by the test is the same as the true sampling distribution $p^*(T)$. Let $T^{(s)} \sim p^*(T)$ be fixed. Due to equality of the two distribution, we have $p^{(s)} := P(T_0 \leq T^{(s)}) = P(T^* \leq T^{(s)})$, for random variables $T_0 \sim p_0(T)$ and $T^* \sim p^*(T)$. Since both, $T^{(s)}, T^* \sim p^*(T)$ come from the same distribution, the sampling distribution of $F^{(s)} := P(T^* \leq T^{(s)})$ is uniform (proven earlier). Accordingly, the sampling distribution of $p^{(s)}$ must be uniform too.

A statistical test producing a uniform p-curve under M_0 is said to be *(well) calibrated* for M_0 .

T-test for homogenous variances

Setting: We have datasets y_j of two groups $j = 1, 2$ with N_j number of normally distributed observations:

$$y_{ji} \sim M_j = \text{normal}(\mu_j, \sigma)$$

for unknown μ_j and σ .

Under the null hypothesis $H_0 : \mu_1 = \mu_2$, the t-test for homogenous variances with test statistic

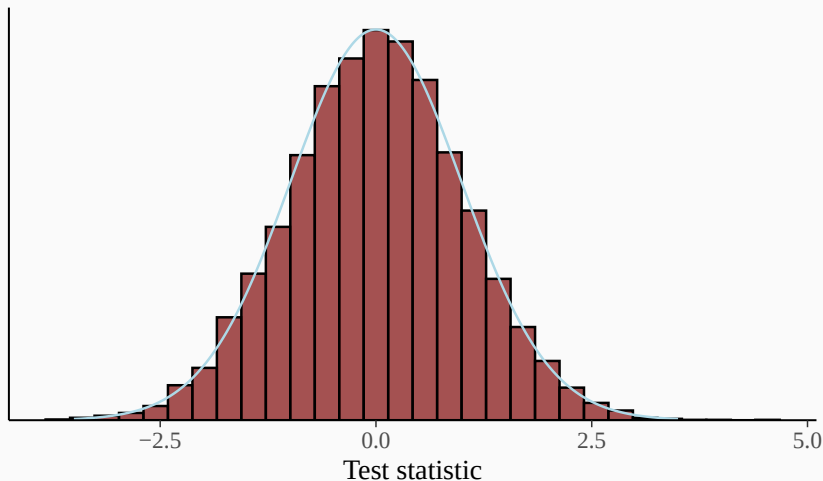
$$t = \frac{\bar{y}_1 - \bar{y}_2}{\hat{\sigma}_p^2 \sqrt{\frac{1}{N_1} + \frac{1}{N_2}}}$$

where

$$\hat{\sigma}_p^2 = \sqrt{\frac{(N_1 - 1)\hat{\sigma}_1^2 + (N_2 - 1)\hat{\sigma}_2^2}{N_1 + N_2 - 2}}$$

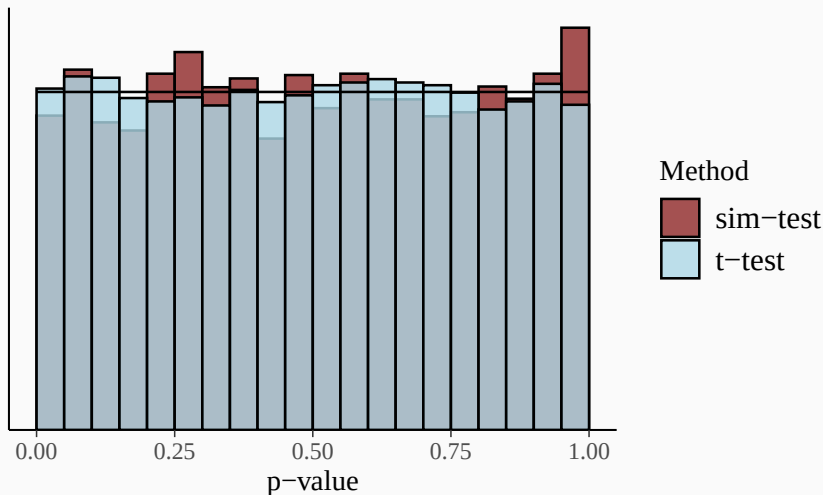
is (well) calibrated.

Sampling distribution of the t-test statistic for normal data



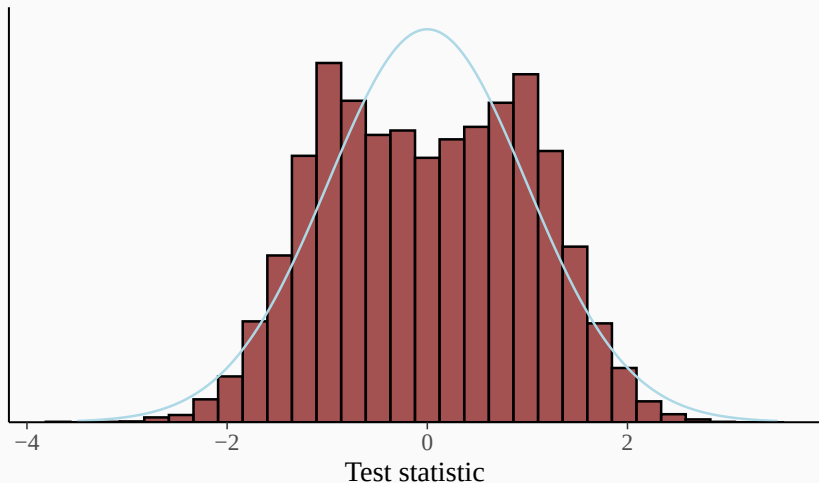
Lightblue: Assumed sampling distribution of the t-test. Red: Histogram of the true sampling distribution.

P-curve of the t-test for normal data



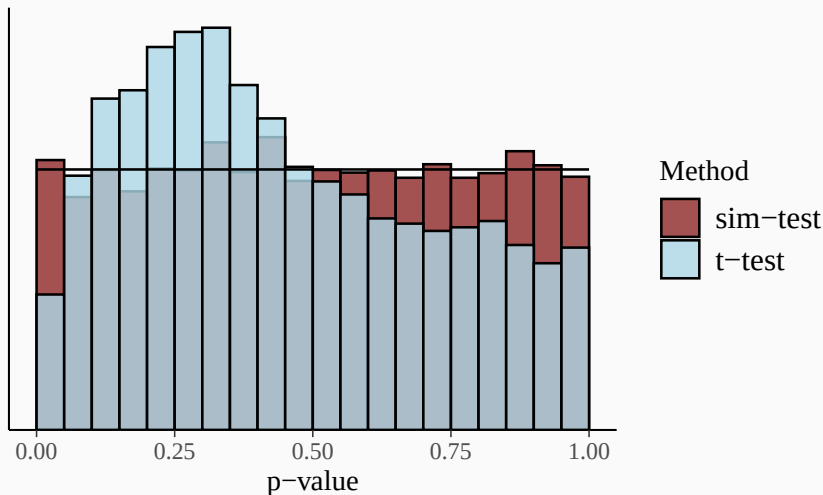
Lightblue: P-curve of the t-test. Red: P-curve of a simulation-based test using $M_0 = M^*$.

Sampling distribution of the t-test statistic for *lognormal* data



Lightblue: Assumed sampling distribution of the t-test. Red: Histogram of the true sampling distribution.

P-curve of the t-test for *lognormal* data



Lightblue: P-curve of the t-test. Red: P-curve of a simulation-based test using $M_0 = M^*$.

Consider a statistical test that is well calibrated under a null hypothesis model M_0 .

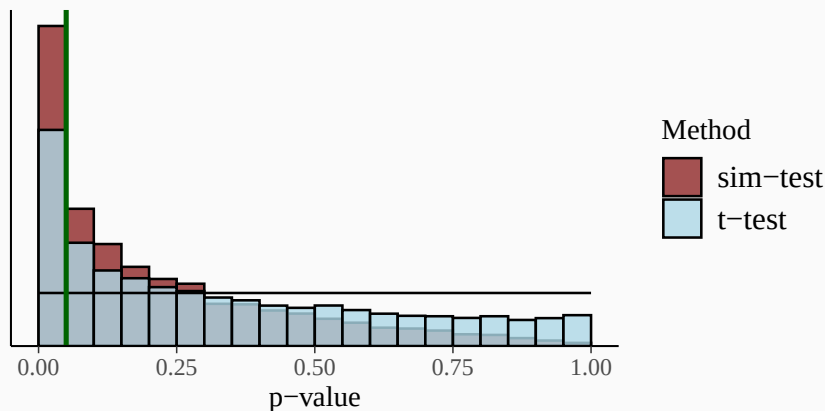
Let $M_1 \neq M_0$ be another model representing the substantive (alternative) hypothesis H_1 .

Fix the nominal type-I error rate α , say, $\alpha = 0.05$.

Then, the probability $P(p^{(s)} \leq \alpha)$ for $p^{(s)}$ obtained from $y^{(s)} \sim M_1$ is called (statistical) power of test under M_1 .

Statistical power for normal data

Consider normal data with $\mu_1 = 2$, $\mu_2 = 2.5$, and $\sigma = 2$.



T-test power = 0.204. Sim-test power = 0.302.

Definition: Let M^* be the true model with true parameter value θ^* , and let α be the nominal Type-I error rate. An interval of the form $[\theta_l(y), \theta_u(y)]$ is a *(well) calibrated* confidence interval for θ^* , if $\theta^* \in [\theta_l(y), \theta_u(y)]$ with probability $1 - \alpha$ for data $y \sim M^*$.

Calibration: CIs for the mean

Standard $1 - \alpha$ CI based on $\bar{y}$:

$$[\bar{y} + Q_{\alpha/2} \text{SE}_{\bar{y}}, \bar{y} + Q_{1-\alpha/2} \text{SE}_{\bar{y}}]$$

where $Q_{\alpha/2}$, $Q_{1-\alpha/2}$ are quantiles of the standard normal distribution and

$$\text{SE}_{\bar{y}} = \frac{\sigma}{\sqrt{N}}$$

Question: Under which conditions is the above CI calibrated for μ ?

Calibration: CIs for the mean

General procedure:

- Simulated S datasets $y^{(s)}$ from $M(\theta^*, \psi)$.
- Compute confidence intervals $\text{CI}^{(s)}(\theta)$.
- Compute the percentage of simulations trials for which $\theta^* \in \text{CI}^{(s)}(\theta)$.

Example: Estimate parameter μ in model M : $y_i \sim \text{normal}(\mu, 15)$ via the empirical mean $\bar{y}$ and corresponding normal-based CI.

Calibration results:

Cal95	Cal80	Cal66	Cal50
0.947	0.798	0.664	0.503

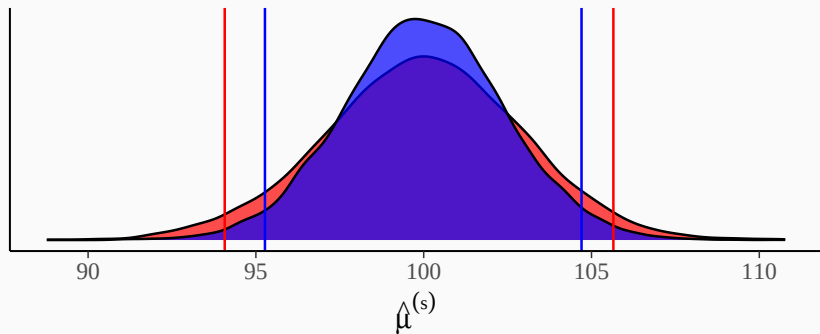
Among two calibrated confidence intervals CI_1 and CI_2 , CI_1 is called *sharper* than CI_2 , if the volume of CI_1 is smaller than the volume CI_2 .

Question: Why can sharpness only be defined on calibrated CIs?

Question: Any idea for two calibrated CIs that are differently sharp?

Sharpness: Mean vs. Median

Goal: Estimate parameter μ in model M : $y_i \sim \text{normal}(\mu, 15)$.



Estimator ■ median ■ mean

The CIs of both mean and median are simulation-based under M , so must both be well calibrated. Yet, the CI of the mean is smaller.

Modeling of simulation results

Often, we have many different simulation conditions to study.

Examples for simulation conditions:

- Different likelihood (e.g., normal vs. lognormal likelihood)
- Different nuisance parameters (e.g., varying σ)
- Different estimators $\hat{\theta}$ (e.g., mean vs. median)
- Different sample sizes N

In order to better understand their interactions, we can model the simulation study data as if it was real data.

Additionally, explicit modeling is relevant if the simulation budget S is small, say, $S < 1000$.

Modeling of simulation results

Example of a simple multilevel model for simulation results via the lme4 R package:

```
library(lme4)
fit <- glmer(
  rmse ~ 1 + like * sigma * N * estimator +
    (1 + estimator | dataset),
  data = simulation_results,
  ...
)
```